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Discrete Random Variables and Probability Distributions

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Probability Distributions for Discrete Random Variables

Probability Distributions for Discrete Random Variables

Probabilities assigned to various outcomes in $\mathcal{S}$ in turn determine probabilities associated with the values of any particular rv X .

The *probability distribution* of X says how the total probability of 1 is distributed among (allocated to) the various possible X values.

Example:

Suppose, for example, that a business has just purchased four laser printers, and let X be the number among these that require service during the warranty period.

Probability Distributions for Discrete Random Variables

Possible X values are then 0, 1, 2, 3, and 4. The probability distribution will tell us how the probability of 1 is subdivided among these five possible values—how much probability is associated with the X value 0, how much is apportioned to the X value 1, and so on.

We will use the following notation for the probabilities in the distribution:

$p(0)$ = the probability of the X value 0 = $P(X = 0)$

$p(1)$ = the probability of the X value 1 = $P(X = 1)$

and so on. In general, $p(x)$ will denote the probability assigned to the value x .

Example 7

The Cal Poly Department of Statistics has **a lab with six computers** reserved for statistics majors.

Let X denote the number of these computers that are in use at a particular time of day.

Suppose that the probability distribution of X is as given in the following table; the first row of the table lists the possible X values and the second row gives the probability of each such value.

x	0	1	2	3	4	5	6
$p(x)$	.05	.10	.15	.25	.20	.15	.10

Example 7

cont'd

We can now use elementary probability properties to **calculate other probabilities of interest**. For example, the probability that **at most 2 computers are in use** is

$$P(X \leq 2) = P(X = 0 \text{ or } 1 \text{ or } 2)$$

$$= p(0) + p(1) + p(2)$$

$$= .05 + .10 + .15$$

$$= .30$$

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Example 7

cont'd

Since the event *at least 3 computers are in use* is complementary to *at most 2 computers are in use*,

$$P(X \geq 3) = 1 - P(X \leq 2)$$

$$= 1 - .30$$

$$= .70$$

which can, of course, also be obtained by adding together probabilities for the values, 3, 4, 5, and 6.

Example 7

cont'd

The probability that **between 2 and 5 computers inclusive** are in use is

$$\begin{aligned}P(2 \leq X \leq 5) &= P(X = 2, 3, 4, \text{ or } 5) \\&= .15 + .25 + .20 + .15 \\&= .75\end{aligned}$$

whereas the probability that the number of computers in use is **strictly between 2 and 5** is

$$\begin{aligned}P(2 < X < 5) &= P(X = 3 \text{ or } 4) \\&= .25 + .20 \\&= .45\end{aligned}$$

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Probability Distributions for Discrete Random Variables

Definition

The **probability distribution** or **probability mass function** (pmf) of a discrete rv is defined for every number x by $p(x) = P(X = x) = P(\text{all } s \in \mathcal{S}: X(s) = x)$.

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Probability Distributions for Discrete Random Variables

In words, for every possible value x of the random variable, the pmf specifies the probability of observing that value when the experiment is performed.

The conditions $p(x) \geq 0$ and $\sum_{\text{all possible } x} p(x) = 1$ are required of any pmf.

The pmf of X in the previous example was simply given in the problem description.

We now consider several examples in which various probability properties are exploited to obtain the desired distribution.

Example 8

Six boxes of components are ready to be shipped by a certain supplier. The number of defective components in each box is as follows:

<i>Box</i>	1	2	3	4	5	6
<i>Number of defectives</i>	0	2	0	1	2	0

One of these boxes is to be randomly selected for shipment to a particular customer.

Let X be the number of defectives in the selected box.

Example 8

$$p(0) = P(X = 0) = P(\text{box 1 or 3 or 6 is sent}) = \frac{3}{6} = .500$$

$$p(1) = P(X = 1) = P(\text{box 4 is sent}) = \frac{1}{6} = .167$$

$$p(2) = P(X = 2) = P(\text{box 2 or 5 is sent}) = \frac{2}{6} = .333$$

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Example 9

Consider whether the next person buying a computer at a certain electronics store buys a laptop or a desktop model. Let

$$p(0) = P(X = 0) = P(\text{next customer purchases a laptop model}) = .8$$

$$p(1) = P(X = 1) = P(\text{next customer purchases a desktop model}) = .2$$

$$p(x) = P(X = x) = 0 \text{ for } x \neq 0 \text{ or } 1$$

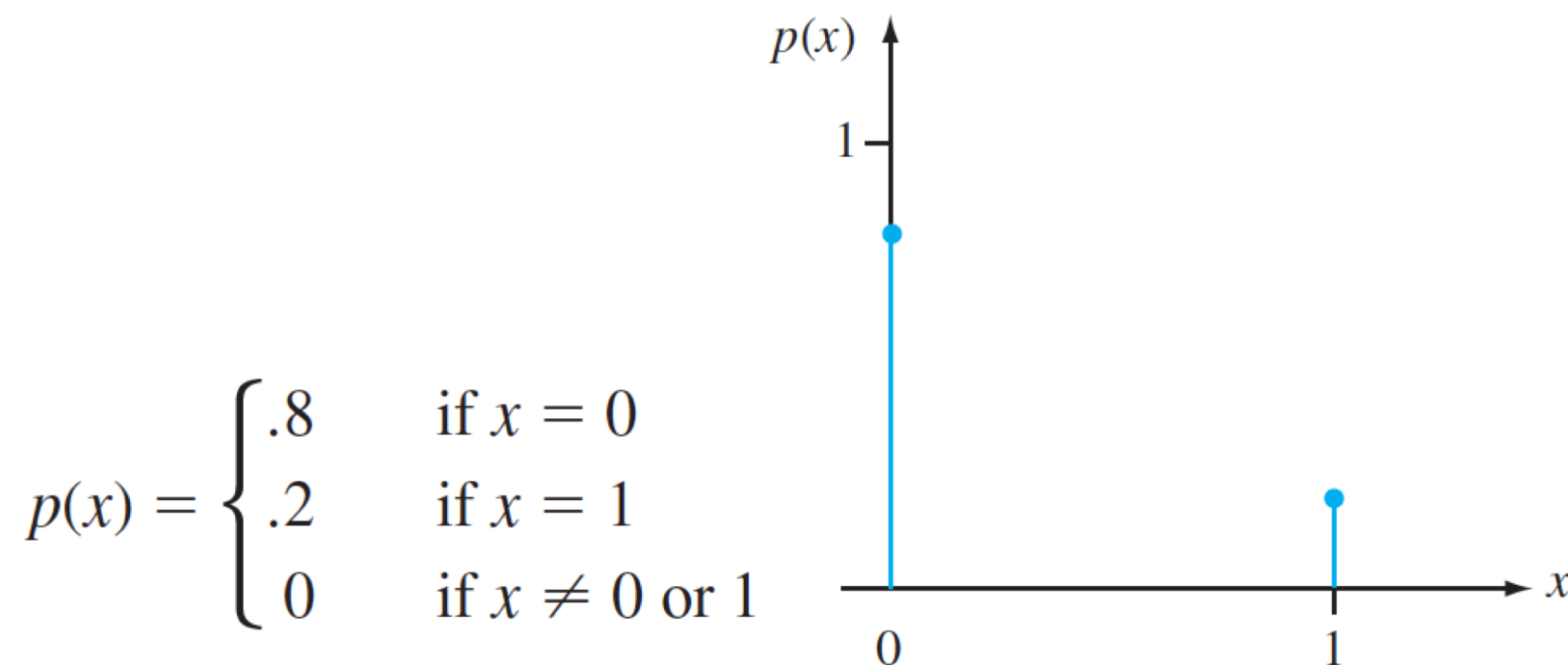
If 20% of all purchasers during that week select a desktop,
We calculate the probability distribution or probability mass function (pmf) of X :

Example 9

$p(0) = P(X = 0) = P(\text{next customer purchases a laptop model}) = .8$

$p(1) = P(X = 1) = P(\text{next customer purchases a desktop model}) = .2$

$p(x) = P(X = x) = 0$ for $x \neq 0$ or 1





A Parameter of a Probability Distribution

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A Parameter of a Probability Distribution

The pmf of the Bernoulli rv X was $p(0) = .8$ and $p(1) = .2$ because 20% of all purchasers selected a desktop computer.

At another store, it may be the case that $p(0) = .9$ and $p(1) = .1$.

More generally, the pmf of any Bernoulli rv can be expressed in the form $p(1) = \alpha$ and $p(0) = 1 - \alpha$, where $0 < \alpha < 1$. Because the pmf depends on the particular value of α we often write $p(x; \alpha)$ rather than just $p(x)$:

$$p(x; \alpha) = \begin{cases} 1 - \alpha & \text{if } x = 0 \\ \alpha & \text{if } x = 1 \\ 0 & \text{otherwise} \end{cases}$$

A Parameter of a Probability Distribution

Then each choice of a in Expression (3.1) yields a different pmf.

Definition

Suppose $p(x)$ depends on a quantity that can be assigned any one of a number of possible values, with each different value determining a different probability distribution. Such a quantity is called a **parameter** of the distribution. The collection of all probability distributions for different values of the parameter is called a **family** of probability distributions.

A Parameter of a Probability Distribution

The quantity α in Expression (3.1) is a parameter. Each different number α between 0 and 1 determines a different member of the Bernoulli family of distributions.

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Example 12

Starting at a fixed time, we observe the gender of each newborn child at a certain hospital until a boy (B) is born.

Let $p = P(B)$, assume that successive births are independent, and define the rv X by x = number of births observed.

Then

$$p(1) = P(X = 1)$$

$$= P(B)$$

$$= p$$

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Example 12

cont'd

$$\begin{aligned}p(2) &= P(X = 2) \\&= P(GB) \\&= P(G) \cdot P(B) \\&= (1 - p)p\end{aligned}$$

and

$$\begin{aligned}p(3) &= P(X = 3) \\&= P(GGB) \\&= P(G) \cdot P(G) \cdot P(B) \\&= (1 - p)^2 p\end{aligned}$$

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Example 12

cont'd

Continuing in this way, a general formula emerges:

$$p(x) = \begin{cases} (1 - p)^{x-1}p & x = 1, 2, 3, \dots \\ 0 & \text{otherwise} \end{cases}$$

The parameter p can assume any value between 0 and 1. Expression (3.2) describes the family of *geometric distributions*.

In the gender example, $p = .51$ might be appropriate, but if we were looking for the first child with Rh-positive blood, then we might have $p = .85$.



The Cumulative Distribution Function

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The Cumulative Distribution Function

For some fixed value x , we often wish to compute the probability that the observed value of X will be at most x .

$$p(x) = \begin{cases} .500 & x = 0 \\ .167 & x = 1 \\ .333 & x = 2 \\ 0 & \text{otherwise} \end{cases}$$

The probability that X is at most 1 is then

$$P(X \leq 1) = p(0) + p(1) = .500 + .167 = .667$$

The Cumulative Distribution Function

In this example, $X \leq 1.5$ if and only if $X \leq 1$, so

$$P(X \leq 1.5) = P(X \leq 1) = .667$$

Similarly,

$$P(X \leq 0) = P(X = 0) = .5, P(X \leq .75) = .5$$

And in fact for any x satisfying $0 \leq x < 1$, $P(X \leq x) = .5$.

The Cumulative Distribution Function

The largest possible X value is 2, so

$$P(X \leq 2) = 1 \quad P(X \leq 3.7) = 1 \quad P(X \leq 20.5) = 1$$

and so on.

Notice that $P(X < 1) < P(X \leq 1)$ since the latter includes the probability of the X value 1, whereas the former does not.

More generally, when X is discrete and x is a possible value of the variable, $P(X < x) < P(X \leq x)$.

The Cumulative Distribution Function

Definition

The **cumulative distribution function** (cdf) $F(x)$ of a discrete rv variable X with pmf $p(x)$ is defined for every number x by

$$F(x) = P(X \leq x) = \sum_{y: y \leq x} p(y) \quad (3.3)$$

For any number x , $F(x)$ is the probability that the observed value of X will be at most x .

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Example 13

A store carries flash drives with either 1 GB, 2 GB, 4 GB, 8 GB, or 16 GB of memory.

The accompanying table gives the distribution of Y = the amount of memory in a purchased drive:

y	1	2	4	8	16
$p(y)$	.05	.10	.35	.40	.10

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Example 13

cont'd

Let's first determine $F(y)$ for each of the five possible values of Y :

$$\begin{aligned} F(1) &= P(Y \leq 1) \\ &= P(Y = 1) \\ &= p(1) \\ &= .05 \end{aligned}$$

$$\begin{aligned} F(2) &= P(Y \leq 2) \\ &= P(Y = 1 \text{ or } 2) \\ &= p(1) + p(2) \\ &= .15 \end{aligned}$$

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Example 13

cont'd

$$\begin{aligned} F(4) &= P(Y \leq 4) \\ &= P(Y = 1 \text{ or } 2 \text{ or } 4) \\ &= p(1) + p(2) + p(4) \\ &= .50 \end{aligned}$$

$$\begin{aligned} F(8) &= P(Y \leq 8) \\ &= p(1) + p(2) + p(4) + p(8) \\ &= .90 \end{aligned}$$

$$\begin{aligned} F(16) &= P(Y \leq 16) \\ &= 1 \end{aligned}$$

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Example 13

cont'd

Now for any other number y , $F(y)$ will equal the value of F at the closest possible value of Y to the left of y . For example,

$$\begin{aligned} F(2.7) &= P(Y \leq 2.7) \\ &= P(Y \leq 2) \\ &= F(2) \\ &= .15 \end{aligned}$$

$$\begin{aligned} F(7.999) &= P(Y \leq 7.999) \\ &= P(Y \leq 4) \\ &= F(4) \\ &= .50 \end{aligned}$$

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Example 13

cont'd

If y is less than 1, $F(y) = 0$ [e.g. $F(.58) = 0$], and if y is at least 16, $F(y) = 1$ [e.g. $F(25) = 1$]. The cdf is thus

$$F(y) = \begin{cases} 0 & y < 1 \\ .05 & 1 \leq y < 2 \\ .15 & 2 \leq y < 4 \\ .50 & 4 \leq y < 8 \\ .90 & 8 \leq y < 16 \\ 1 & 16 \leq y \end{cases}$$

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Example 13

cont'd

A graph of this cdf is shown in Figure 3.5.

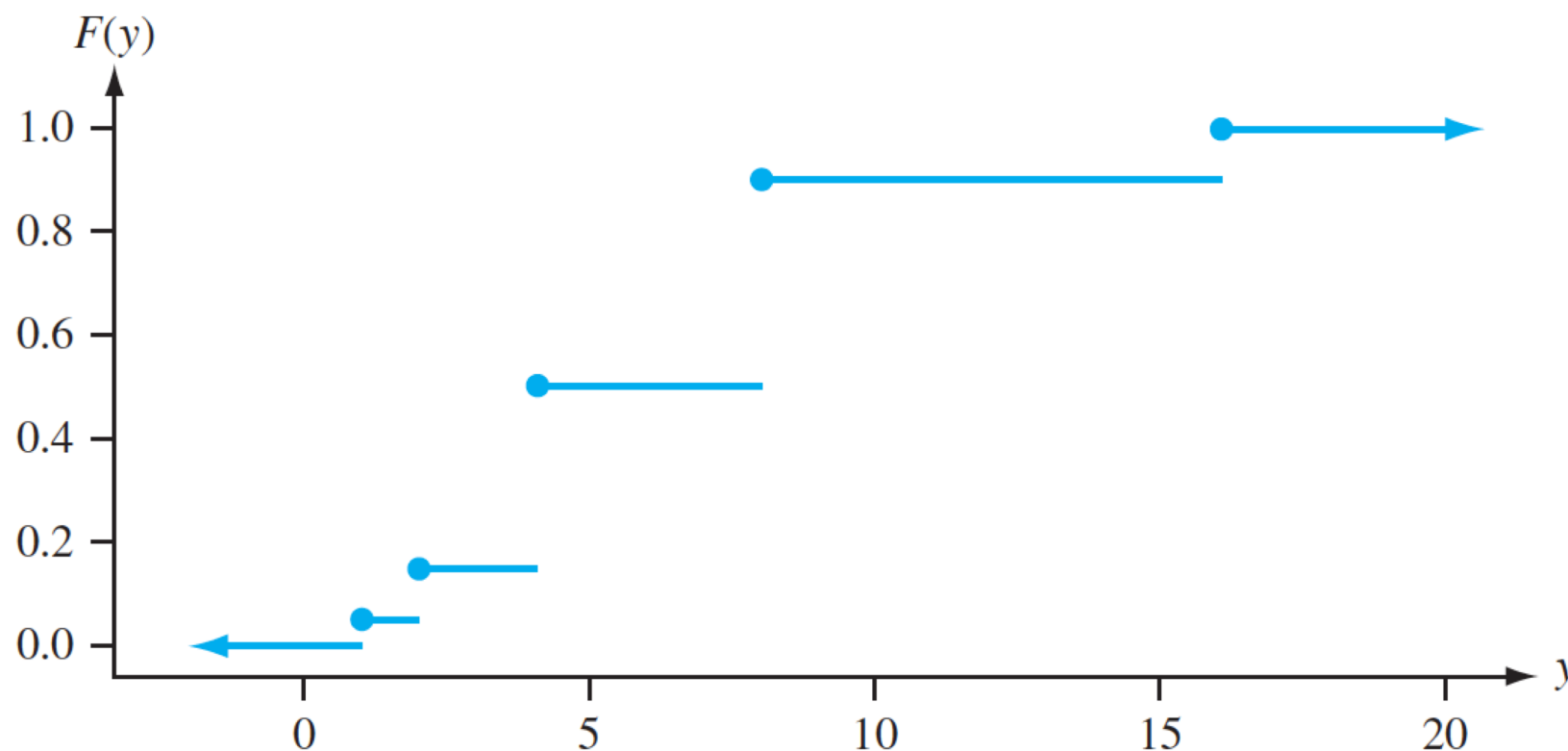


Figure 3.5 A graph of the cdf of Example 3.13

The Cumulative Distribution Function

For X a discrete rv, the graph of $F(x)$ will have a jump at every possible value of X and will be flat between possible values. Such a graph is called a **step function**.

Proposition

For any two numbers a and b with $a \leq b$,

$$P(a \leq X \leq b) = F(b) - F(a-)$$

where “ $a-$ ” represents the largest possible X value that is strictly less than a .

The Cumulative Distribution Function

In particular, if the only possible values are integers and if a and b are integers, then

$$\begin{aligned} P(a \leq X \leq b) &= P(X = a \text{ or } a + 1 \text{ or } \dots \text{ or } b) \\ &= F(b) - F(a - 1) \end{aligned}$$

Taking $a = b$ yields $P(X = a) = F(a) - F(a - 1)$ in this case.

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The Cumulative Distribution Function

PROPOSITION

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where “ $a-$ ” represents the largest possible X value that is strictly less than a . In particular, if the only possible values are integers and if a and b are integers, then

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Taking $a = b$ yields $P(X = a) = F(a) - F(a - 1)$ in this case.

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The Cumulative Distribution Function

The reason for subtracting $F(a-)$ rather than $F(a)$ is that we want to include $P(X = a)$. $F(b) - F(a-)$ gives $P(a < X \leq b)$.

This proposition will be used extensively when computing binomial and Poisson probabilities in Sections 3.4 and 3.6.

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Example 15

Let X = the number of days of sick leave taken by a randomly selected employee of a large company during a particular year.

If the maximum number of allowable sick days per year is 14, possible values of X are 0, 1, . . . , 14.

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Example 15

cont'd

With $F(0) = .58$, $F(1) = .72$, $F(2) = .76$, $F(3) = .81$,
 $F(4) = .88$, $F(5) = .94$,

$$P(2 \leq X \leq 5)$$

$$P(2 \leq X \leq 5) = P(X = 2, 3, 4, \text{ or } 5)$$

$$= F(5) - F(1)$$

$$= .22$$

$$P(X = 3)$$

$$P(X = 3) = F(3) - F(2)$$

$$= .05$$

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